Investigation: BAHEMA SA (CVM code 701), fiscal year 2015, as it was on 2016-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Emp. Adm. Part. - Sem Setor Principal (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

Document used

2015 DFP, version 1, filed on 2016-03-22; origin: open data; type: individual; original scale: MIL.
Versions known on the date: 1 (2016-03-22).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=54269

Possible manipulation (engine v1, study 1)

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 11.383 (index)
Components: X1 = 0.927; X2 = 0.109; X3 = −0.073; X4 = 5.182 (indices)
Percentile in sector and year: 87 (n = 49)
The company's own history: median 10.636; Tukey range (1.5 × IQR): from 4.419 to 16.645; value inside the range (index; 5 fiscal years)
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 1/9 (7 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 0
- Change in ROA (ΔROA): 0
- Cash above earnings (ACCRUAL): 0
- Change in leverage (ΔLEVER): 0
- Change in current ratio (ΔLIQUID): 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): —
- Change in asset turnover (ΔTURN): 0

- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement
- No long-term debt in either year: the change in leverage (ΔLEVER) is zero and the signal is 0 (project choice; the article only defines fell = 1, rose = 0, p. 8)

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

What was known on the date

- DFP 2010, v1: 2011-03-04
- DFP 2011, v1: 2012-03-16
- DFP 2012, v1: 2013-04-01
- DFP 2013, v1: 2014-03-31
- DFP 2014, v1: 2015-03-28
- FCA 2015, v1: 2015-03-28
- FCA 2015, v2: 2015-05-04
- DFP 2015, v1: 2016-03-22

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 3 failed.

- Cash flow closing balance = cash and cash equivalents on the balance sheet (ÚLTIMO): difference above the tolerance of R$2,000
- Cash flow closing balance = cash and cash equivalents on the balance sheet (PENÚLTIMO): difference above the tolerance of R$2,000
- Cash flow opening balance = cash and cash equivalents on the prior balance sheet (ÚLTIMO): difference above the tolerance of R$2,000

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
no opinion excerpt was found in the text
Auditor in the 2013 FCA: Audilink & Cia.Ltda..
Auditor in the 2014 FCA: Audilink & Cia.Ltda..
Auditor in the 2015 FCA: BKS Auditores.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2015: Audilink & Cia.Ltda. -> BKS Auditores. The outgoing firm audited 5 consecutive fiscal years (2010 to 2014), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 36 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E).
Generated at 2026-11-11T06:36:21+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: BIOMA EDUCAÇÃO S.A..